THE NOVA RESEARCH CURRICULUM

Mathematics

Grade 1 · Term 1

Student edition

Term 1 outcomes: MA-1.1 (part) numbers 0–20, MA-1.7 shapes and patterns, MA-1.6 comparing. Six lessons a week. Every lesson moves from **objects** (counters, sticks, beads) to **pictures** to **numerals**.

Lesson notes

MA-1.1 (part) Counts, reads and writes numbers 0–20 (Weeks 1–8)

1. **Count real things:** count counters, fingers, pencils; touch each one as you count.
2. **Match:** match a group of objects to its numeral card.
3. **Ten frame:** show numbers on a 10-frame; 11–20 as "ten and some more" using two frames.
4. **Before, after, between (Weeks 6–8):** a number line on the floor; children jump to the number before or after.
5. **Write:** trace and write numerals; 0–9 first, then 10–20.

Watch out: counting an object twice; writing 2, 3, 5, 7 backwards; reading 12 as 21. Use the 10-frame to show 12 is ten and two.

MA-1.7 Names shapes; continues a pattern (Weeks 9–10)

1. **Feel:** a bag of shapes; children find a shape by touch and name it.
2. **Hunt:** find squares, rectangles, circles and triangles in the room or at home.
3. **Patterns:** continue AB and ABC patterns with shapes, colours and claps.

Watch out: calling a square a rectangle is not wrong (a square is a special rectangle) but in Grade 1 use the simpler names; a triangle turned upside down is still a triangle.

MA-1.6 Compares length, height and weight (Weeks 11–12)

1. **Compare two:** two pencils, two children, two bags: longer/shorter, taller/shorter, heavier/lighter.
2. **Order three:** put three objects in order.
3. **Balance:** use a hanger balance or hands to feel heavier and lighter.

Watch out: thinking bigger always means heavier (a balloon and a stone).

▶ **Try it on NovaSchool** (open **Simulations** on your dashboard):

- **Ten frames: count to 20** — Tap + to add a counter and − to take one away. When the first frame is full, it makes a ten.

Workbook pages

Page	Outcome	Activity
1–4	MA-1.1	Count and circle the number (1–10), one page per 2–3 numbers.
5	MA-1.1	Draw dots in the 10-frame to show the number.

Page	Outcome	Activity
6–8	MA-1.1	Count and write 11–20 using two 10-frames.
9	MA-1.1	Fill the missing numbers on a number line 0–20.
10	MA-1.1	Before, after, between: __ 7 __ ; 12 __ 14.
11	MA-1.7	Colour circles red, squares blue, triangles green, rectangles yellow.
12	MA-1.7	Draw what comes next in the pattern.
13	MA-1.6	Tick the taller, longer and heavier picture.

Question bank

MA-1.1 (numbers 0–20)

#	Question	Your answer
1	[5 apples] How many apples?	4 / 5 / 6
2	[8 stars] How many stars?	8 / 9 / 7
3	Which number comes after 6?	5 / 7 / 8
4	Which number comes before 10?	9 / 11 / 8
5	Which number is between 13 and 15?	12 / 14 / 16
6	[one full 10-frame and 2 more dots] What number is this?	12 / 21 / 10
7	17 is bigger than 13. True or false.	True / False
8	[numeral cards 3, 9, 5] Which is the biggest number?	3 / 5 / 9
9	0 means none. True or false.	True / False
10	[a full 10-frame and another full 10-frame] What number is this?	10 / 20 / 2
11	Count on: 15, 16, 17, __	18 / 19 / 20
12	11 is ten and one. True or false.	True / False

MA-1.7

#	Question	Your answer
1	[3 shapes] Which one is a circle?	[circle] / [square] / [triangle]
2	A triangle has 3 sides. True or false.	True / False
3	[picture: a coin] What shape is it?	circle / triangle / square
4	[picture: a door] What shape is it?	circle / rectangle / triangle
5	A square has 4 equal sides. True or false.	True / False
6	[red, blue, red, blue, __] What comes next?	red / blue / green
7	[circle, square, circle, square, __] What comes next?	square / circle / triangle

#	Question	Your answer
8	[upside-down triangle] Is this still a triangle? True or false.	True / False
9	[star, star, moon, star, star, __] What comes next?	star / moon / sun
10	A circle has corners. True or false.	True / False

MA-1.6

#	Question	Your answer
1	[two pencils] Which pencil is longer?	[long pencil] / [short pencil]
2	[a giraffe and a cat] Which is taller?	giraffe / cat
3	[an elephant and a mouse] Which is heavier?	elephant / mouse
4	A feather is heavier than a brick. True or false.	True / False
5	[three ropes] Which rope is the shortest?	[long] / [middle] / [short]
6	[balance: left side down] Which side is heavier?	left / right
7	A big balloon is always heavier than a small stone. True or false.	True / False
8	[a child and an adult] Who is shorter?	child / adult
9	[a spoon and a ladle] Which is longer?	spoon / ladle
10	Taller means it goes higher up. True or false.	True / False